21/01/2026

Code-A

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, *Ph.*+91-1244168300

MM : 720

NCERT Booster Program for NEET-2026_NBTS-04A

Time : 180 Min.

Topics Covered:
Physics: Oscillations, Waves, Electric Charges & Fields, Electrostatic Potential and Capacitance, Current Electricity, Moving Charges and Magnetism, Magnetism and Matter
Chemistry: Solutions, Electrochemistry, The d- and f-Block Elements, Coordination Compounds
Botany: Plant Growth and Development, Sexual reproduction in flowering plants, Principles of Inheritance and Variation
Zoology: Animal Kingdom, Structural Organisation in Animals: Animal Morphology of Frog and Cockroach, Human Reproduction

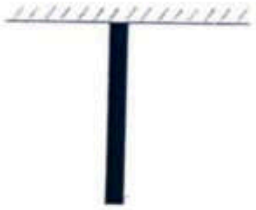
General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

CC-075 CC-075 CC-075

Vivek Krishna
Aakash Medical | IIT-JEE | Foundations

PHYSICS

1. A uniform string of length L is suspended from the ceiling. At $t = 0$ a transverse wave is propagated from its lower end. At the same instant a particle is dropped from the ceiling. Distance from the ceiling at which they pass each other is



(1) $\frac{L}{2}$

(2) $\frac{L}{3}$

(3) $\frac{2L}{3}$

(4) $\frac{L}{4}$

CC-075 CC-075

2. A wire of length 1 m and mass density 4×10^3 kg/m^3 is stretched between two clamps and subjected to an extension of 3.6×10^{-4} m. If Young's modulus of the wire is 4×10^{10} N/m^2 then the frequency of the first overtone of the transverse vibration in the wire is

(1) 30 Hz

(2) 60 Hz

(3) 50 Hz

(4) 40 Hz

3. The equation of a wave disturbance propagating in +ve x-direction is given by $y = \frac{3}{2+x^2}$ at $t = 0$ and $y = \frac{3}{2+(x-1)^2}$ at $t = 1$ s where x and y are in meter assuming that the shape of the wave disturbance does not change during the propagation, the speed of the wave will be

(1) 1 m/s

(2) 4 m/s

(3) 3 m/s

(4) 2 m/s

4. The vibrations of a string of length 300 cm fixed at both ends are represented by $y = 2\sin\left(\frac{\pi x}{30}\right)\cos(90\pi t)$ (All unit are in cgs)

(1) Total number of nodes is 11

(2) Total number of antinodes is 10

(3) Amplitude of a point on string at x = 5 cm is 1 cm

(4) All of these

5. Let there be a spherically symmetric charge distribution with charge density varying as $\rho(r) = \rho_0\left[\frac{5}{4} - \frac{r}{R}\right]$ upto $r = R$ and $\rho(r) = 0$ for $r > R$, where r is the distance from origin.The electric field at a distance r ($r < R$) from the origin is given by.

(1) $\frac{4\pi\rho_0 r}{3\varepsilon_0}\left[\frac{5}{3} - \frac{r}{R}\right]$

(2) $\frac{\rho_0 r}{4\varepsilon_0}\left[\frac{5}{3} - \frac{r}{R}\right]$

(3) $\frac{4\rho_0 r}{3\varepsilon_0}\left[\frac{5}{3} - \frac{r}{R}\right]$

(4) $\frac{\rho_0 r}{3\varepsilon_0}\left[\frac{5}{3} - \frac{r}{R}\right]$

CC-075

6. For a uniformly positively charged ring of radius a on its axis at a small distance "x", a point charge $-q$, is released ($x << a$). If λ is charge density on ring, mass of $-q$ is m, its time period of oscillation is

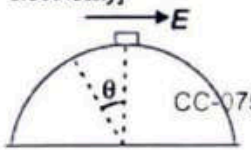
(1) $T = 2\pi\varepsilon_0\sqrt{\frac{2m}{\lambda q}}$

(2) $T = 2\pi a\sqrt{\frac{2\varepsilon_0 m}{\lambda q}}$

(3) $T = 2\pi\varepsilon_0 a\sqrt{\frac{2m}{\lambda q}}$

(4) $T = 2\pi\sqrt{\frac{2\varepsilon_0 m}{\lambda q}}$

7. A negatively charged small disc is released on the top of a fixed hemispherical frictionless inverted bowl in presence of a uniform horizontal electric field. If the disc leaves the bowl at an angle $\theta = \tan^{-1}\frac{3}{4}$. The ratio of gravitational and electrostatic forces on the disc is [Bowl does not interact electrically]



CC-075

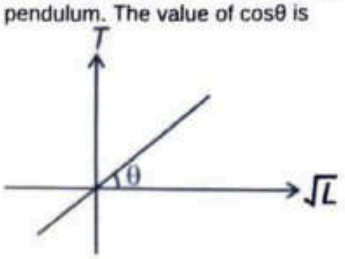
(1) $\frac{5}{3}$

(2) $\frac{9}{2}$

(3) $\frac{15}{2}$

(4) $\frac{11}{3}$

8. In a simple pendulum experiment, a graph is plotted between time period T and $\sqrt{L}$, where L is length of the pendulum. The value of cosθ is



(1) $\frac{2\pi}{\sqrt{9}}$

(2) $\frac{4\pi}{\sqrt{9}}$

(3) $\cos\left[\tan^{-1}\left(\frac{2\pi}{\sqrt{9}}\right)\right]$

(4) $\cos\left[\sin^{-1}\left(\frac{2\pi}{\sqrt{9}}\right)\right]$

9. A circular disc has a tiny hole in it, at a distance z from its center. Its mass is M and radius $R(R > z)$. A horizontal shaft is passed through the hole and held fixed so that the disc can freely swing in the vertical plane. For small disturbance, the disc performs SHM. The value of z for which time period is minimum is

CC-075

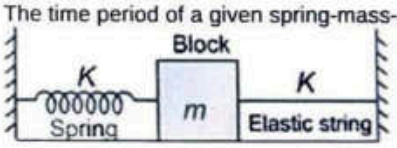
(1) $\frac{R}{\sqrt{2}}$

(2) $\frac{R}{2}$

(3) $\frac{R}{4}$

(4) $\frac{R}{\sqrt{3}}$

10. The time period of a given spring-mass-string system is



(1) $2\pi\sqrt{\frac{m}{2K}}$

(2) $2\pi\sqrt{\frac{m}{K}}$

(3) $(\sqrt{2}+1)\pi\sqrt{\frac{m}{2K}}$

(4) $(\sqrt{2}+1)\pi\sqrt{\frac{m}{K}}$

CC-075

Vivek Krishna Aakash Medical/IIT-JEE Foundations

11. Match Column -I with Column - II (All the symbols have their usual meaning).

Column - I	Column - II
(A) Ampere's circuital law	(i) $\left(\vec{M} \times \vec{B}\right)$
(B) Force per unit length between parallel current carrying wires	(ii) $q\left(\vec{E} + \vec{v} \times \vec{B}\right)$
(C) Lorentz force	(iii) $\frac{\mu_0 i_1 i_2}{2\pi d}$
(D) Torque on a circular current carrying loop placed in uniform magnetic field	(iv) $\oint \vec{B} . d\vec{l} = \mu_0 i_{enclosed}$

(1) A → iv, B → iii, C → i, D → ii

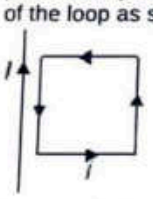
(2) A → iv, B → iii, C → ii, D → i

(3) A → ii, B → i, C → iii, D → iv

(4) A → i, B → iii, C → ii, D → iv

12. A current carrying rectangular loop is kept in a magnetic field produced by a straight current carrying wire kept in the plane of the loop as shown. The loop will experience a net force

CC-075



(1) Towards left

(2) Towards right

(3) Upwards

(4) Zero

13. Dimensional formula of magnetic flux is

(1) $[ML T^{-3} A^{2}]$

(2) $[ML^{2} T^{-2} A^{-1}]$

(3) $[ML^{-2} T^{2} A]$

(4) $[ML^{-2} TA^{-1}]$

14. The area of B-H curve measures

(1) The energy loss per unit area of the specimen

(2) The energy loss per unit volume per cycle of the specimen

CC-075

(3) The energy loss by the specimen

(4) The rate of energy loss by the specimen

15. Magnetic field due to a bar magnet on its equatorial line will be

(1) Parallel to magnetic moment

(2) Anti-parallel to magnetic moment

(3) Perpendicular to magnetic moment

(4) At an angle 30° with direction of magnetic moment

16. A magnet of magnetic moment M is cut into two equal part along its length. The new magnetic moment of each part of the magnet is

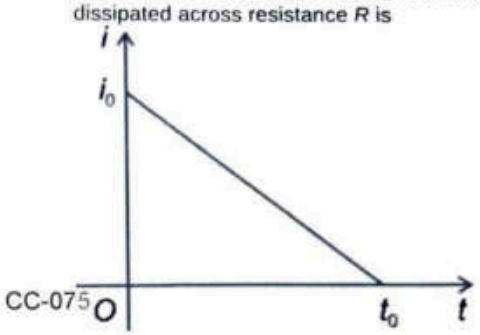
(1) M

(2) $\frac{M}{4}$

(3) $2M$

(4) $\frac{M}{2}$

17. The figure given below shows the variation of current i with time t. If the resistance of the conductor is R then total heat dissipated across resistance R is



CC-075 CC-075

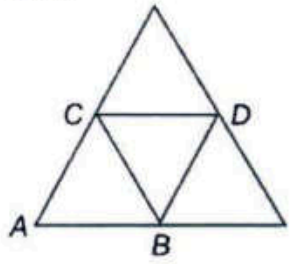
(1) $\frac{i_0^2 R t_0}{4}$

(2) $\frac{i_0^2 R t_0}{2}$

(3) $i_0^2 R t_0$

(4) $\frac{i_0^2 R t_0}{3}$

18. In the diagram resistance of each section is R. Equivalent resistance across the points A and B as shown in the figure will be



(1) $\frac{7}{11}R$

(2) $\frac{11}{18}R$

CC-075 CC-075

(3) $\frac{18}{11}R$

(4) $\frac{11}{7}R$

Vivek Kanna Aakash Medical|IIT-JEE|Foundations

19. If n, e, τ and m are representing electron density, charge, relaxation time and mass of an electron respectively, then the resistance of a wire of length l and cross-sectional area A is given by

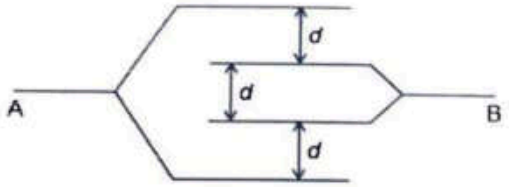
(1) $\frac{ml}{ne^2\tau A}$

(2) $\frac{m\tau A}{ne^2 l}$

(3) $\frac{ne^2\tau A}{ml}$

(4) $\frac{ne^2 A}{m\tau l}$

20. If area of each plate is A, then net capacitance of the given arrangement is



(1) $\frac{\varepsilon_0 A}{d}$

(2) $\frac{3\varepsilon_0 A}{d}$

(3) $\frac{\varepsilon_0 A}{2d}$

(4) $\frac{2\varepsilon_0 A}{d}$

CC-075

21. When the separation between two charges is increased, the electric potential energy of the system of charges

(1) Increases

(2) Decreases

(3) Remains same

(4) May increase or decrease

22. Dielectric constant for a metal is

(1) Zero

(2) Infinite

(3) 1

(4) 10

23. Consider the following statements and choose the correct option.

(a) The velocity of simple harmonic oscillator is maximum at mean position.

(b) Acceleration of simple harmonic oscillator is maximum of extreme position.

(c) The acceleration and displacement of a body are proportional to each other but opposite in direction.

(1) Only (a) and (b) are correct

(2) Only (b) and (c) are correct

(3) Only (a) and (c) are correct

(4) All (a), (b) and (c) are correct

CC-075

24. The angular velocity and amplitude of a simple pendulum are 'ω' and A respectively. At a displacement 'X' iF its kinetic energy is 'K' and potential energy is 'U', then the ratio of 'K' to U is

(1) $\left(\frac{A^2}{X^2} - 1\right)$

(2) $\left(\frac{A^2}{X^2\omega^2} - 1\right)$

(3) $\left(\frac{A^2}{X^2} - \frac{X^2}{\omega^2}\right)$

(4) $\left(\frac{A^2}{\omega^2} - 1\right)$

25. Two masses m and $2m$ are suspended together by a massless spring of force constant 'K'. When the masses are in equilibrium, $2m$ is removed without disturbing the system. The maximum velocity of oscillation of mass m will be

(1) $g\sqrt{\frac{m}{K}}$

(2) $2g\sqrt{\frac{m}{K}}$

(3) $g\sqrt{\frac{2m}{K}}$

(4) $g\sqrt{\frac{m}{2K}}$

CC-075

26. A progressive wave of frequency 400 Hz is travelling with a speed of 350 m/s. A compressional maxima appears at a place at a given instant. The minimum time interval after which a rarefaction maxima occurs at the same place.

(1) $\frac{1}{200}$ s

(2) $\frac{1}{800}$ s

(3) $\frac{1}{400}$ s

(4) $\frac{1}{100}$ s

27. Two tuning forks when sounded together produce 6 beats in 3 s. The time interval between consecutive maximum and minimum intensities of sound is

(1) $\frac{1}{6}$ s

(2) $\frac{1}{8}$ s

(3) $\frac{1}{4}$ s

(4) $\frac{1}{2}$ s

CC-075

Vivek Kr. Arya
Aakash
Medical | IIT-JEE | Foundations

28. Given below are two statements, one is the assertion (A) and the other is reason (R).

Assertion (A): The phase difference between two medium particles having path difference $\frac{\lambda}{2}$ is π.

Reason (R) : If two medium particles have path difference equal to Δx, then phase difference between them is given by, $\Delta\phi = \frac{2\pi}{\lambda} \cdot \Delta x$.

In the light of above statements, select the most appropriate option.

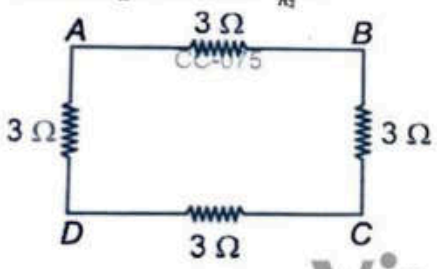
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

29. Four identical resistances are joined as shown in figure. The equivalent resistance between D and C is R_1 and between D and B is R_2. The ratio of $\frac{R_1}{R_2}$ is



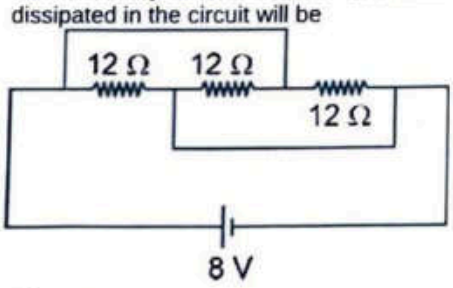
(1) 2 : 5

(2) 2 : 3

(3) 4 : 5

(4) 3 : 4

30. Three resistors each having resistance 12 Ω are connected with a battery of emf 8 V as shown in figure. The power dissipated in the circuit will be



(1) 16 W

(2) 8 W

(3) 12 W

(4) 4 W

31. The current in the circuit varies with time 't' as $I = t + t^2$. Where I is in ampere and t is in second. The average current flow through section of conductor between $t = 1$ s to $t = 3$ s will be

(1) $\frac{20}{3}$ A

(2) $\frac{19}{3}$ A

(3) $\frac{7}{3}$ A

(4) $\frac{11}{3}$ A

32. In meter bridge experiment when the resistance in gaps are interchanged, the balance point is increased by 15 cm. The ratio of resistances will be

(1) $\frac{23}{17}$

(2) $\frac{12}{7}$

(3) $\frac{13}{11}$

(4) $\frac{14}{3}$

33. Read the following statements.

Statement A: Dia-magnetic substances acquire feeble magnetism opposite to direction of magnetic field applied on it.

Statement B: Dia-magnetic substance obeys Curie law.

In the light of above statements mark the **correct** option.

(1) Both statement A and B are correct

(2) Both statement A and B are incorrect

(3) Statement A is correct and Statement B is incorrect

(4) Statement A is incorrect and Statement B is correct

34. A bar magnet of length 20 cm and pole strength 3 Am makes an angle 37° with uniform magnetic field of induction 50 T. The torque acting on it will be

(1) 12 Nm

(2) 18 Nm

(3) 9 Nm

(4) 16 Nm

35. The mass of an iron rod is 160 g and its magnetic moment is 10 A m^2. If the density of iron is 8 g/cm^3. Then the value of intensity of magnetisation in SI unit will be

(1) 2×10^6

(2) 5×10^5

(3) 2×10^5

(4) 5×10^6

CC-075 CC-075 CC-075 CC-075
Vivek Krishna Aakash Medical|IIT-JEE|Foundations

36. A rectangular coil of length *a*, breadth *b* and number of turns *N* carrying current *i* is placed in uniform magnetic field *B* with its area vector parallel to the lines of magnetic field. The work done in rotating the coil through an angle of 180° is

(1) *iNabB*

(2) 2*iNabB*

(3) Zero

(4) $\frac{iNabB}{2}$

37. A charged particle enters a magnetic field at right angles to the field. The field exists for a length equal to 1.5 times the radius of circular path of particle. The particle will be deviated from its path by

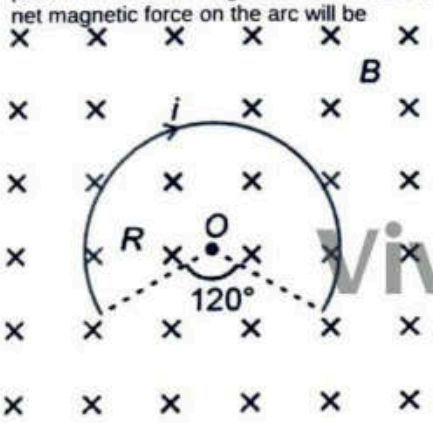
(1) 90°

(2) 180°

(3) 60°

(4) $\sin^{-1}\left(\frac{1}{3}\right)$

38. A current carrying circular arc of radius R having current *i*, placed in uniform magnetic field *B* as shown in figure. The net magnetic force on the arc will be



(1) *BiR*

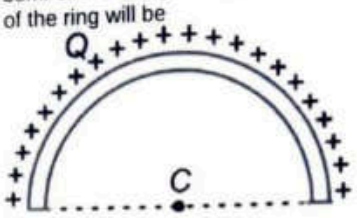
(2) $\frac{\sqrt{3}BiR}{2}$

(3) $\frac{BiR}{2}$

(4) $\sqrt{3}BiR$

CC-075

39. A charge *Q* is distributed uniformly along the periphery of semi-circular ring of radius *R*. The electric field at the centre of the ring will be



(1) $\frac{Q}{2\pi\varepsilon_0 R^2}$

(2) $\frac{Q}{2\pi^2\varepsilon_0 R^2}$

(3) $\frac{Q}{4\pi\varepsilon_0 R^2}$

(4) $\frac{Q}{4\pi^2\varepsilon_0 R^2}$

40. On an isolated conductor of non-uniform curvature, the charge,

CC-075

(1) Has the greatest concentration on the part of greatest radius of curvature

(2) Has the greatest concentration on the parts of least radius of curvature

(3) Is distributed uniformly on the curved surface

(4) Is distributed uniformly over its volume

41. A positive charge 3*Q* is placed on a conducting spherical shell with inner radius *R* and outer radius 2*R*. A point charge *Q* is placed inside the shell. The surface charge density of outer surface of sphere will be

(1) $\frac{Q}{4\pi R^2}$

(2) $\frac{Q}{\pi R^2}$

(3) $\frac{Q}{8\pi R^2}$

(4) $\frac{3Q}{16\pi R^2}$

42. The two equilateral triangular plate of side *a* are placed parallel to each other at distance *d* in vaccum. The capacitance of the configuration will be

CC-075

(1) $\frac{\sqrt{3}a^2\varepsilon_0}{2d}$

(2) $\frac{\sqrt{3}a^2\varepsilon_0}{4d}$

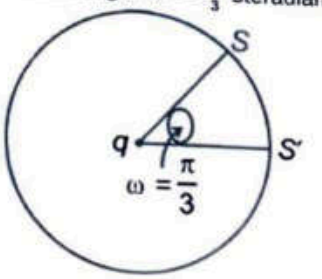
(3) $\frac{a^2\varepsilon_0}{d}$

(4) $\frac{\sqrt{3}a^2\varepsilon_0}{d}$

CC-075

Vivek Rajput Aakash Medical|IIT-JEE|Foundation

43. A charge q is placed at the centre of sphere as shown in figure. The flux passing through surface SS' which subtends a solid angle $\omega = \frac{\pi}{3}$ steradian at centre will be



(1) $\frac{q}{6\varepsilon_0}$

(2) $\frac{q}{4\varepsilon_0}$

(3) $\frac{q}{12\varepsilon_0}$

(4) $\frac{q}{9\varepsilon_0}$

44. In uniform electric field, $\vec{E} = 3E_0\hat{i} + 2E_0\hat{j}$, where E_0 is a constant in the region of space and at (0, 0) the electric potential is zero, then electric potential at (X_0, X_0) will be

CC-075

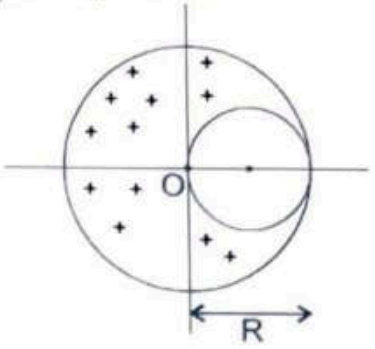
(1) $-2E_0X_0$

(2) $-5E_0X_0$

(3) $-3E_0X_0$

(4) Zero

45. A solid sphere of radius R is uniformly charged with charge density ρ in its volume. A spherical cavity of radius $\frac{R}{2}$ is made in the sphere as shown then electric potential at the centre of sphere will be



(1) $\frac{3\rho R^2}{8\varepsilon_0}$

(2) $\frac{5\rho R^2}{6\varepsilon_0}$

CC-075

(3) $\frac{5\rho R^2}{12\varepsilon_0}$

(4) $\frac{3\rho R^2}{4\varepsilon_0}$

CC-075

Vivek Krishna
Aakash
Medical|IIT-JEE|Foundations

CHEMISTRY

46. Mass percent of solute in 0.1 m aqueous glucose solution will be

(1) 2.22%

(2) 1.76%

(3) 1.05%

(4) 2.86%

CC-075

47. If Henry's Law constant (K_H) values in water for four gases A, B, C and D are 30.2, 1.32, 1.07×10^{-4} and 0.27 kbar respectively at the same temperature, then the gas which has the maximum solubility in water is

(1) A

(2) B

(3) C

(4) D

48. A solution is made by two volatile components $A\left(P^*_A = 100 \text{ mm Hg}\right)$ and $B\left(P^*_B = 200 \text{ mm Hg}\right)$. Find out the mole fraction of A in vapour phase if solution is made by mixing equal moles of A and B.

CC-075

(1) $\frac{1}{2}$

(2) $\frac{1}{3}$

(3) $\frac{2}{3}$

(4) $\frac{1}{4}$

CC-075

49. Given below are two statements:
Statement-I: 95% ethanol in water by volume, makes an azeotropic mixture.
Statement-II: Maximum boiling azeotropes are solutions formed by solutions with positive deviation under azeotropic conditions.
In the light of above statements, choose the **correct** option

(1) Both statement-I and II are true

(2) Both statement-I and II are false

(3) Statement-I is false but statement II is true

(4) Statement-I is true but statement II is false

50. Which amongst the following aqueous solutions will have minimum freezing point? [assume 100% ionisation of electrolyte]

(1) 0.005 m Glucose

(2) 0.1 m $Al_2(SO_4)_3$

(3) 0.01 m $MgSO_4$

(4) 0.2 m KCl

51. The mass of urea to be dissolved in 100 g of water (K_b = 0.52 K kg mol^{-1}) to prepare a solution having boiling point 101°C, is

(1) 10.26 g

(2) 4.2 g

(3) 11.54 g

(4) 14.66 g

52. Given below are two statements: one is labelled as **assertion (A)** and the other is labelled as **reason (R)**
Assertion (A): Conductivity always decreases with decrease in concentration for both weak and strong electrolytes.
Reason (R): The number of ions per unit volume that carry the charge in a solution decreases on dilution.
In the light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

CC-075

53. If at 25°C molar conductance at infinite dilution of CH_3COOH is 500 Scm^2mol^{-1} and the degree of dissociation of CH_3COOH is 0.1 for 0.1 molar aqueous solution, then the molar conductance (in Scm^2 mol^{-1}) of CH_3COOH at same concentration is

(1) 510

(2) 250

(3) 50

(4) 25

54. Match List I with List II.

	List I (Conversion)		**List II (Number of Faraday required)**
A.	1 mol of H_2O to O_2	I.	3F
B.	1 mol of MnO_4^- to Mn^{2+}	II.	2F
C.	1.5 mol of Ca from molten $CaCl_2$	III.	1F
D.	1 mol of FeO to Fe_2O_3	IV.	5F

CC-075

Choose the **correct** answer from the options given below:

(1) A-II, B-IV, C-I, D-III

(2) A-III, B-IV, C-I, D-II

(3) A-II, B-III, C-I, D-IV

(4) A-III, B-IV, C-II, D-I

55. Given below are two statements, one is labelled as assertion (A) and the other is labelled as reason (R).
Assertion (A): On electrolysis of aqueous NaCl solution using inert electrodes, pH of the solution increases.
Reason (R) : On electrolysis of aqueous NaCl, water gets reduced at cathode and concentration of OH^- ion increases in solution.
In the light of above statements, select the most appropriate option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

CC-075

CC-075

Vivek Kumar Aakash Medical/IIT-JEE/Foundations ८

56. Consider the following statements.
(a) The electrolyte used in mercury cell is a paste of KOH and ZnO.
(b) In lead storage battery, 38% solution of sulphuric acid is used as an electrolyte.
(c) In Leclanche cell the space between the electrodes is filled by a moist paste of ammonium chloride and zinc chloride.
The correct statement(s) is/are

(1) (a) and (b) only

(2) (a) and (c) only

(3) (a), (b) and (c)

(4) (c) only

57. At 298 K, the standard electrode potentials of Mg^{2+}/Mg, Ni^{2+}/Ni, Cu^{2+}/Cu and Au^{3+}/Au are –2.36 V, –0.25 V, 0.34 V and 1.40 V respectively
On the basis of standard electrode potential, predict which of the following reactions can occur?

(1) $Ni(s) + Mg^{2+}(aq) \rightarrow Ni^{2+}(aq) + Mg(s)$

(2) $Cu(s) + Ni^{2+}(aq) \rightarrow Cu^{2+}(aq) + Ni(s)$

(3) $2Au^{3+}(aq) + 3Mg(s) \rightarrow 2Au(s) + 3Mg^{2+}(aq)$

(4) $3Ni^{2+}(aq) + 2Au(s) \rightarrow 2Au^{3+}(aq) + 3Ni(s)$

58. Consider the following statements
(a) Scandium shows only +3 oxidation state
(b) Mn^{2+} has least negative hydration enthalpy among 3d series elements
(c) Zinc has highest density among 3d series elements
(d) CrO is amphoteric and Cr_2O_3 is acidic in nature
(e) Brass is an alloy of copper and zinc
The correct statements are

(1) (a), (b), (c), (d) and (e)

(2) (a), (b) and (e) only

(3) (a), (b) and (c) only

(4) (b), (c) and (e) only

59. Consider the following two statements.
Statement I: Ce(IV) is a good analytical reagent.
Statement II: Lanthanoids liberate hydrogen gas from dilute acids.
In the light of above statements, choose the correct option.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

60. Aqueous solution of which ion is colourless?

(1) Co^{2+}

(2) Cu^{2+}

(3) Sc^{3+}

(4) V^{3+}

61. Alkaline oxidative fusion of MnO_2 produces (A) which undergoes electrolytic oxidation in alkaline solution giving (B) as the product. (A) and (B) respectively are

(1) MnO_4^- and MnO_4^{2-}

(2) Mn_2O_7 and Mn^{2+}

(3) MnO_4^{2-} and MnO_4^-

(4) Mn^{2+} and Mn_2O_7

62. The correct electronic configuration of Gadolinium and Lutetium respectively are

(1) $[Xe]4f^7\ 5d^1\ 6s^2$ and $[Xe]\ 4f^{14}\ 5d^1\ 6s^2$

(2) $[Xe]4f^8\ 6s^2$ and $[Xe]\ 4f^{14}\ 6s^2$

(3) $[Xe]4f^9\ 6s^1$ and $[Xe]\ 4f^{13}\ 5d^2\ 6s^2$

(4) $[Xe]4f^{13}\ 5d^1\ 6s^2$ and $[Xe]\ 4f^7\ 5d^1\ 6s^2$

63. Most basic compound among the following is

(1) $La(OH)_3$

(2) $Eu(OH)_3$

(3) $Er(OH)_3$

(4) $Lu(OH)_3$

64. Given below are two statements:
Statement I: Type of isomerism exhibited by the compounds $[Co(NH_3)_5NO_2]Cl_2$ and $[Co(NH_3)_5ONO]Cl_2$ is linkage isomerism.
Statement II: The coordination number of cobalt in both the complexes, $[Co(NH_3)_5NO_2]Cl_2$ and $[Co(NH_3)_5ONO]Cl_2$ is 6.
In the light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

65. Which of the following set of species acts as an ambidentate ligand?

(1) Ethane-1, 2-diamine and $C_2O_4^{2-}$

(2) $EDTA^{4-}$ and $C_2O_4^{2-}$

(3) NO_2^- and SCN^-

(4) Ethane-1, 2-diamine and NO_2^-

CC-075 CC-075 CC-075 CC-075 CC-075 CC-075
Vivek Kasana Aakash Medical/IIT-JEE/Foundations

66. The correct electronic configuration of central metal atom 'Fe' in $[Fe(CN)_6]^{4-}$ based on crystal field theory is

(1) $t_{2g}^5 e_g^1$

(2) $t_{2g}^4 e_g^2$

(3) $t_{2g}^6 e_g^0$

(4) $t_{2g}^3 e_g^3$

67. Shape of $[Ni(CN)_4]^{2-}$ is

(1) Tetrahedral

(2) Square planar

(3) Trigonal pyramidal

(4) See saw

68. The correct order of increasing ligand field strength is

(1) $CO < H_2O < NH_3 < Cl^-$

(2) $Cl^- < H_2O < NH_3 < CO$

(3) $H_2O < NH_3 < Cl^- < CO$

(4) $Cl^- < NH_3 < H_2O < CO$

69. For which of the following complexes, the C-O bond length is maximum?

(1) $[Mn(CO)_6]^+$

(2) $[Ni(CO)_4]$

(3) $[Cr(CO)_6]$

(4) $[V(CO)_6]^-$

70. The correct IUPAC name of $K_2[PdCl_4]$ is

(1) Potassium tetrachloropalladium (II)

(2) Dipotassium tetrachloridopalladate (IV)

(3) Potassium tetrachloridopalladate (II)

(4) Potassium tetrachloropalladium (IV)

71. Which one is not equal to zero for an ideal solution?

(1) ΔH_{mix}

(2) ΔV_{mix}

(3) $\Delta P = P_{observed} - P_{calculated\ by\ Raoult's\ law}$

(4) ΔG_{mix}

72. Which one of the following electrolytes has the same value of van't Hoff's factor (i) as that of $Ca_3(PO_4)_2$? (assume 100% ionization)

(1) K_2SO_4

(2) $Al(NO_3)_3$

(3) $K_4[Fe(CN)_6]$

(4) $K_3[Fe(CN)_6]$

73. The osmotic pressure of a solution by dissolving 17.4 mg K_2SO_4 in 1.5 litre of water at 27°C is
(assume that it is completely dissociated)

(1) 4.926×10^{-3} atm

(2) 3.762×10^{-3} atm

(3) 5.932×10^{-3} atm

(4) 2.532×10^{-3} atm

74. The depression in the freezing point of water when 8.8 g of $CH_3CH_2CH_2COOH$ is added to 250 g of water at T K will be
(Given that $K_a = 2.5 \times 10^{-6}$, $K_f = 1.86$ K Kg mol^{-1})

(1) 0.62 K

(2) 1.025 K

(3) 0.89 K

(4) 0.75 K

75. Given below are two statements.
Statement I: When isotonic solutions are separated by semipermeable membrane no osmosis occurs between them.
Statement II: The osmotic pressure method is widely used to determine molar mosses of macromolecules.
In the light of above statements, choose the **correct** answer.

(1) Statement I is correct but statement II is incorrect

(2) Both statement I and statement II are incorrect

(3) Both statement I and statement II are correct

(4) Statement I is incorrect but statement II is correct

76. The standard electrode potential (E°) values of Mg^{2+}/Mg, Ag^+/Ag, K^+/K and Zn^{2+}/Zn are –2.36 V, 0.80 V, –2.93 V and –0.76 V, respectively. The correct decreasing order of reducing power of the metals is

(1) Mg > K > Ag > Zn

(2) Ag > Zn > Mg > K

(3) K > Mg > Zn > Ag

(4) K > Mg > Ag > Zn

CC-075 CC-075 CC-075 CC-075 CC-075
Vivek Kumar Aakash Medical|IIT-JEE|Foundations

77. Match the list-I with list-II

	List-I (Ion)		List-II (Limiting Molar Conductivity/S cm^2 mol^{-1})
a.	H^+	(i)	106
b.	Ca^{2+}	(ii)	199.1
c.	OH^-	(iii)	349.6
d.	Mg^{2+}	(iv)	119

The correct match is

(1) a-iii, b-iv, c-ii, d-i

(2) a-iii, b-iv, c-i, d-ii

(3) a-iv, b-iii, c-ii, d-i

(4) a-ii, b-i, c-iii, d-iv

78. The equilibrium constant (K_a) of weak acid of c molar concentration can be given as [Where Λ_m = molar conductivity, Λ_m° = limiting molar conductivity]

CC-075

(1) $K_a = \frac{c\Lambda_m}{\Lambda_m(\Lambda_m^\circ - \Lambda_m)}$

(2) $K_a = \frac{c\Lambda_m^2}{\Lambda_m^\circ(\Lambda_m - \Lambda_m^\circ)}$

(3) $K_a = \frac{c\Lambda_m^2}{\Lambda_m^\circ(\Lambda_m^\circ - \Lambda_m)}$

(4) $K_a = \frac{c\Lambda_m^2}{\Lambda_m(\Lambda_m^\circ - \Lambda_m)}$

79. On electrolysis of an aqueous solution of $CuCl_2$ using platinum (Pt) electrode, the product obtained at cathode will be

(1) Hydrogen gas

(2) Oxygen gas

(3) Chlorine gas

(4) Copper metal

80. Given below are two statements.

Statement I: In rusting of iron atmospheric oxygen is reduce to water at cathode

Statement II: Electrochemical cells are extensively used for determining the pH of solution, solubility product and equilibrium constant.

CC-075

In the light of above statements, choose the **correct** answer.

(1) Statement I is correct but statement II is incorrect

(2) Both statement I and statement II are incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are correct

81. The transition elements which do not exhibit variable oxidation state?

(a) Sc (b) Fe

(c) Mn (d) Zn

(e) Co

Choose the correct option.

(1) (a) and (d) only

(2) (a), (b) and (c) only

(3) (a), (d) and (e) only

(4) (d) and (e) only

82. The E° value of the Mn^{3+}/Mn^{2+} couple is more positive than that of Fe^{3+}/Fe^{2+} due to change of

(1) d^5 to d^4 configuration

(2) d^5 to d^2 configuration

(3) d^3 to d^5 configuration

(4) d^4 to d^5 configuration

CC-075

83. Consider the following statements

(a) Lanthanum, gadolinium and lutetium have abnormally low value of the third ionisation enthalpy.

(b) All the lanthanoids are silvery white soft metals and tarnish rapidly in air

(c) Lanthanoids are good conductors of heat and electricity.

Choose the correct statements.

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a), (b) and (c)

(4) (a) and (c) only

84. The pair of lanthanoid ions which is diamagnetic in nature is

(1) Ce^{4+} and Lu^{3+}

(2) Sm^{3+} and Yb^{2+}

(3) Gd^{3+} and Eu^{3+}

(4) Pm^{3+} and Sm^{3+}

CC-075 CC-075

Vivek Krishna
Aakash Medical | IIT-JEE | Foundations

85. Match the list-I with list-II.

	List-I (Catalyst)		List-II (Process)
a.	$PdCl_2$	(i)	Oxidation of SO_2 in the manufacture of sulphuric acid
b.	Nickel complexes	(ii)	Polymerisation of ethylene
c.	V_2O_5	(iii)	Oxidation of ethene to ethanal
d.	$TiCl_4 + Al(CH_3)_3$	(iv)	Polymerisation of alkynes

Choose the correct match.

(1) a-iii, b-iv, c-i, d-ii

(2) a-iii, b-iv, c-ii, d-i

(3) a-iv, b-iii, c-ii, d-i

(4) a-ii, b-iii, c-i, d-iv

86. The existence of two different coloured complexes with the composition of $[Co(NH_3)_4Cl_2]^+$ is due to

(1) Linkage isomerism

(2) Geometrical isomerism

(3) Coordination isomerism

(4) Optical isomerism

87. An excess of $AgNO_3$ is added to 200 ml of a 0.02 M solution of chloropentaamminecobalt(III)chloride. The number of moles of AgCl precipitated would be

(1) 0.004

(2) 0.008

(3) 0.04

(4) 0.08

88. Which of these statements about $[Co(NH_3)_6]^{3+}$ is true?

(1) $[Co(NH_3)_6]^{3+}$ has no unpaired electrons and will be in a high-spin configuration

(2) $[Co(NH_3)_6]^{3+}$ has no unpaired electrons and will be in a low-spin configuration

(3) $[Co(NH_3)_6]^{3+}$ has four unpaired electrons and will be in a low-spin configuration

(4) $[Co(NH_3)_6]^{3+}$ has four unpaired electrons and will be in a high-spin configuration

89. Which one of the following is expected to exhibit optical isomerism? (en = ethylenediamine)

(1) cis– $[Pt(NH_3)_2Cl_2]$

(2) trans– $[Pt(NH_3)_2Cl_2]$

(3) cis– $[CoCl_2(en)_2]$

(4) trans– $[CoCl_2(en)_2]$

90. Given below are two statements.

Statement I: $[Co(H_2O)_6]^{3+}$ is a homoleptic complex whereas $[Co(NH_3)_4Cl_2]^+$ is a heteroleptic complex.

Statement II: Complex $[Co(H_2O)_6]^{3+}$ has only one kind of ligands but $[Co(NH_3)_4Cl_2]^+$ has more than one kind of ligands.

In the light of above statements, choose the **correct** answer.

(1) Statement I is correct but statement II is incorrect

(2) Both statement I and statement II are incorrect

(3) Both statement I and statement II are correct

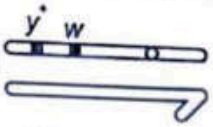
(4) Statement I is incorrect but statement II is correct

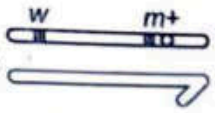
CC-075 CC-075 CC-075 CC-075 CC-075

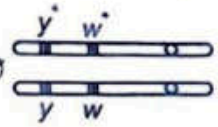
Vivek Raisha Aakash Medical|IIT-JEE|Foundation

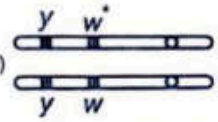
BOTANY

91. In Morgan's experiment on *Drosophila*, considering its eye colour (w/w^+), body colour (y/y^+) and wing size (m/m^+), which option does **not** represent recombinant type?

(1) 

(2) 

(3) 

(4) 

CC-075

92. The scientist who isolated auxin from the tips of coleoptiles of oat seedlings was

(1) F.W. Went

(2) Charles Darwin and Francis Darwin

(3) F. Skoog

(4) E. Kurosawa

93. Select the **correctly** matched pair from the following options

(1)	$r = \frac{L_t + L_0}{t}$	–	Absolute growth rate
(2)	Efficiency index	–	Growth rate
(3)	$W_1 = W_0 + e^{rt}$	–	Exponential growth
(4)	Larkspur	–	Habitat dependent heterophylly

(1) (1)

(2) (2)

(3) (3)

(4) (4)

94. Growth in plants is

(1) Closed and limited

(2) Open but limited

CC-075

(3) Closed and indeterminate

(4) Open and indeterminate

95. A hormone which promotes female flowers in cucumber is

(1) Gibberellin

(2) Cytokinin

(3) Auxin

(4) Ethylene

96. Miller et al identified and crystallised the cytokinesis promoting active substance known as

(1) Auxin

(2) Kinetin

(3) ABA

(4) Gibberellin

97. Cells/tissues arising out of the same meristem have different structures at maturity. This phenomenon is readily observed in plants and is only possible if

(1) Growth is open ended

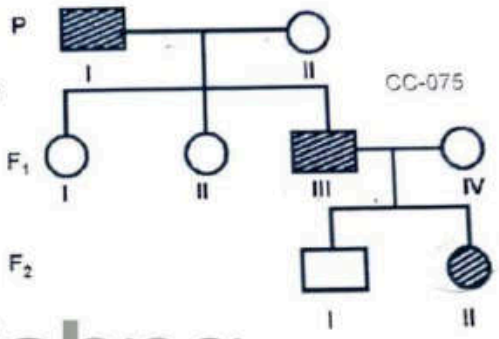
(2) Differentiation is determinate

(3) Growth is indeterminate

(4) Differentiation is open ended

98. Study the given pedigree chart of a family for a particular trait in which the shaded ones represent the trait.

CC-075



CC-075

Which of the following conclusions can be drawn from the chart?

(I) The trait may be autosomal recessive

(II) The disorders, like colour blindness, can be the example for such pedigree chart

(III) The individual represented by F_1 (IV) can never be homozygous.

(IV) The trait under study cannot be X-linked dominant.

(1) All (I), (II), (III) and (IV)

(2) (I) and (IV) only

(3) (IV) only

(4) (I), (II) and (IV) only

CC-075

CC-075

Vivek Vishna Aakash Medical/IIT-JEE

99. In an experiment on artificial hybridisation, pollen from a plant (AaBb) are dusted on stigma of a plant (AAbb). Which of the following options correctly depicts the genotype of zygote and PEN formed in an embryo sac after double fertilisation? Choose the correct option.

	PEN	Zygote
(1)	AAaBbb	Aabb
(2)	AAAbbb	AaBb
(3)	AAabbb	Aabb
(4)	AAABbb	Aabb

(1) (1)

(2) (2)

(3) (3)

(4) (4)

100. In which of the following pairs of genetic disorders, number of autosomes remains unchanged?

(1) Turner's syndrome and Down's syndrome

(2) Klinefelter's syndrome and Down's syndrome

(3) Down's syndrome and Thalassemia

(4) Turner's syndrome and Klinefelter's syndrome

101. Select the **odd** one out w.r.t. post-fertilisation events in flowering plants.

(1) Endosperm development

(2) Embryo development

(3) Maturation of ovule into seed

(4) Formation of embryo sac

102. How many meiotic divisions are required for the formation of 100 egg cells in a typical angiospermic plant?

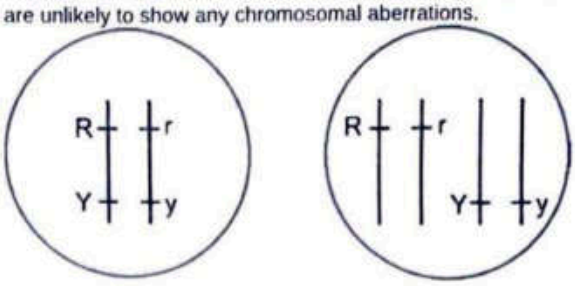
(1) 25

(2) 50

(3) 100

(4) 20

103. The given diagrams depict two allelic pairs within gamete mother cells that show normal chromosomal disjunction and are unlikely to show any chromosomal aberrations.



(A) (B)

Read the following statements regarding the figures given above.

a. Both cells are diploid and would follow the law of segregation.

b. Both cells can produce four different genotypes of gametes.

c. Only one of the cells could ensure independent assortment.

d. Genes R and Y share a linkage group in only one of the cells.

How many of the above given statements are **correct**?

(1) One

(2) Three

(3) Two

(4) Four

104. Identify the **incorrect** statement for apomixis.

(1) Apomictic embryos are genetically identical to each other.

(2) Apomixis is genetically controlled phenomenon

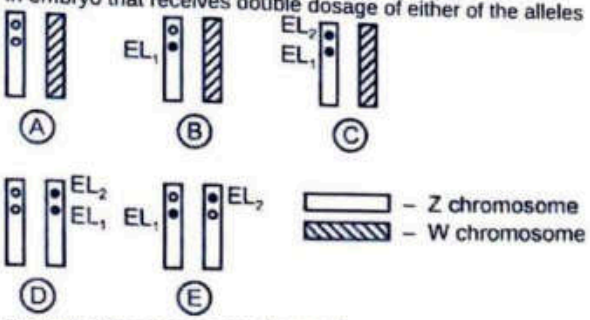
(3) Apomictic embryo can not be diploid as they are not produced after fertilisation

(4) Apomictic embryos are found in *Citrus* fruits.

3124
4244
3443
43244

CC-075 CC-075 CC-075 CC-075 CC-075
Vivek Kasana Aakash Medical/IIT-JEE Foundations

105. In poultry farming, males are needed in bulk to meet consumption demands of an area. Two non-allelic sex-linked recessive mutations El_1 and El_2 are discovered to be lethal in embryo that receives double dosage of either of the alleles



The following crosses are observed
(i) A × D
(ii) B × D
(iii) C × E
(iv) C × D
Which of the following options shows the cross that will result in least percentage of surviving male progeny?

(1) cross iv
(2) cross iii
(3) cross ii
(4) cross i

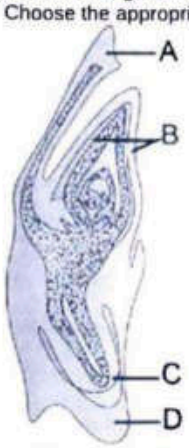
CC-075

106. Fruit flies, *Drosophila melanogaster* were found very suitable for studies on experimental genetics because they

(1) Do not grow on synthetic medium in laboratory conditions
(2) Have no clear differentiation of sexes
(3) Take a very long time to complete their life -cycle
(4) Have many types of hereditary variations that can be seen with low power microscopes

CC-075

107. Observe the given diagram carefully.
Choose the appropriate option regarding A, B, C and D.



	A	B	C	D
(1)	Single developed cotyledon	Contains equal number of chromosomes per cell as compared to that of aleurone layer	Elongates to form main root system of this plant	Undifferentiated sheath that envelops the radicle and root cap
(2)	Observed in seeds such as castor and orchids	Hollow foliar structure that is devoid of leaf primordia	Responsible for the formation of primary root	Made of mature cells
(3)	Its level of attachment on embryo axis demarcates the epicotyl	Encloses the embryonal shoot apex or plumule	Protected by root cap	Has ploidy identical to the main nutritive tissue within a fertilised embryo sac
(4)	Placed laterally to the embryonal axis	Post seed germination, it can exhibit tropic movement	Represents the lower end of hypocotyl	Cannot be observed in plants like mustard and sunflower

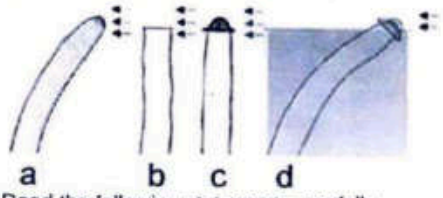
(1) (1)
(2) (2)
(3) (3)
(4) (4)

CC-075

CC-075

Vivek Keshri
Aakash Medical|IIT-JEE|Foundations

108. The given diagram represents different cases of the effect of unilateral illumination on oat coleoptiles.



Read the following statements carefully.

A. Presence of an opaque cover on coleoptile tip negates the impact of unilateral illumination on the transmittable influence and hence prevents bending.

B. A decapitated coleoptile shows tropic response to the same degree as one without decapitation.

C. Removal of the coleoptile tip simply shifts the site of transmittable influence to the stump

D. A coleoptile with a transparent cap has same tropic response as one without it.

Which of the above conclusions can not be drawn from the diagram?

(1) All except A

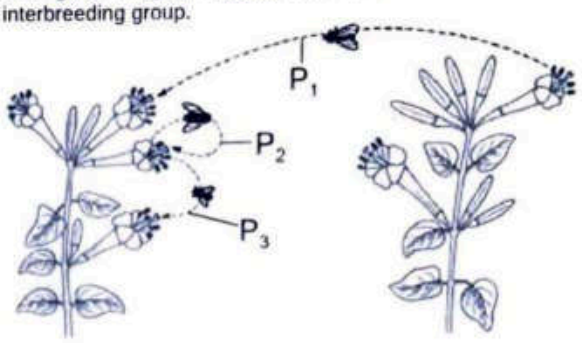
(2) B and C only

(3) A, B and C only

(4) A and D only

CC-075

109. The given diagram depicts two plants taken from an interbreeding group.



Read the following statements based on the above diagram carefully.

(A) Neither P_2 nor P_3 would help in bringing genetic variation.

(B) Only P_1 can bring compatible pollen if the species shows self-incompatibility.

(C) Pollinating agent receives identical floral rewards from P_2 and P_3 but not from P_1

(D) If kept in isolation both plants can survive but future progenies may suffer increasing probability of recessive abnormalities.

(E) Plants A and B cannot be maize plants but can be castor plants.

How many of the above statements are **true**?

(1) Five

(2) Four

(3) Three

(4) Two

CC-075

110. Persistent nucellus in some seeds is called

(1) Endosperm

(2) Pericarp

(3) Perisperm

(4) Epiblast

111. All of the following ensure prevention of inbreeding depression, **except**

(1) Cleistogamy

(2) Dioecy

(3) Xenogamy

(4) Self incompatibility

CC-075

112. An ovary has many ovules in all of the following, **except**

(1) Papaya

(2) Water melon

(3) Orchids

(4) Mango

Vivek Aakash Medical|IIT-JEE|Foundation

113. A couple in which both, mother and father are carriers of sickle cell anemia gives birth to two pairs of monozygotic twins by two separate pregnancies. Monozygotic twins are a result of splitting of one zygote into two cells and further proceeding to develop into two embryos. What is the probability of both sets of twins to be boys as well as carriers for sickle cell disease?

(1) $\frac{1}{4}$

(2) $\frac{1}{8}$

(3) $\frac{1}{16}$

(4) $\frac{1}{32}$

114. Which type of sex-determination system is present in honey bees?

(1) XY – XX type

(2) Haplodiploid type

(3) ZW – ZO type

(4) ZZ – ZW type

CC-075

115. Given below are some statements regarding growth,
(A) Growth can be expressed as increase in both number as well as size of the cell.
(B) Increase in protoplasm that is the principal consequence of growth at cellular level is difficult to measure directly.
(C) Growth of pollen tube and leaf is measured in term of increase in length and surface area respectively.
(D) Growth is generally measured by increase in a variety of parameters which is more or less proportional to the increase in protoplasm.
Based on the given statements choose the **correct** option.

(1) All the statements are true and (D) is the correct reason for (B)

(2) (B) and (D) are true and (B) is the correct reason for (D)

(3) (A) and (B) are true and (A) is the correct reason for (B)

(4) (A) and (C) are true and (C) is the correct reason for (A)

116. In angiosperms, triple fusion is required for the formation of

(1) Fruit wall

(2) Seed coat

(3) Embryo sac

(4) Endosperm

117. The following statements are given for ABO blood grouping in human beings. Identify the **incorrect** one.

CC-075

(1) The gene *I* has three (multiple) alleles I^A, I^B and *i*.

(2) The alleles I^A and I^B each exhibit codominance when present with *i* and are responsible for a unique AB phenotype.

(3) A total of six genotypes regarding human ABO blood types are present in the population.

(4) Characters like ABO blood grouping can only be studied when population is considered.

118. Which of the following is **not** a similarity between gene mutation and chromosomal aberration?

(1) Both can affect the nucleotide sequence within the DNA molecule

(2) Both can change the ploidy of the organism

(3) Both can alter the expression of a protein

(4) Both have a chance of not resulting in disorder

119. Read the following statements and state them as true (T) and false (F).
(A) The lateral buds of plant post decapitation do not require external application of cytokinins in order to grow.
(B) Gibberellic acid has the ability to cause an increase in length of axis *i.e.*, bolting in plants with rosette habit such as beet, cabbage and sugarcane.
(C) Natural ethylene is synthesised in ripening fruits whereas natural cytokinins are synthesised in juvenile fruit.
(D) Ethylene is used to initiate flowering and for synchronising fruit set in pineapple.
Choose the **correct** option.

CC-075 CC-075

	A	B	C	D
(1)	T	F	T	F
(2)	T	F	T	T
(3)	F	T	T	T
(4)	F	T	F	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

120. Read the following statements carefully.
(A) Better adaptive strategies for dispersal to new habitats.
(B) Plants no longer depend on water in order to form seeds.
(C) Young seedlings are nourished by reserve food within seeds.
(D) Hard seed coat provides protection to young embryo.
(E) Maintenance of characters and prevention of segregation in progeny.
If in an angiosperm both fertilised and unfertilised ovules give rise to seeds, which of the above advantages would be provided by both of the seeds?

(1) All except E

(2) B, D and E only

(3) B, C and D only

(4) A and C only

CC-075 CC-075

Vivek Krishna Aakash Medical|IIT-JEE|Foundations

121. Read the following statements, state **true** (T) or **false** (F) for them and select the **correct** option.
(a) Cleistogamous flowers prevent geitonogamy as well as xenogamy.
(b) Double fertilisation occurs in all spermatophytes.
(c) In pea family, the single cotyledon is called scutellum.
(d) Seeds of castor and coconut retain a part of endosperm during embryo development.

	(a)	(b)	(c)	(d)
(1)	T	F	T	F
(2)	T	T	F	F
(3)	T	F	F	T
(4)	F	T	T	F

(1) (1)
(2) (2)
(3) (3)
(4) (4)

122. In a plant, the zygote undergoes mitosis with a cell cycle completion time of 5 minutes. It initially exhibits geometric growth for 3 mitotic generations, later shifting to arithmetic growth for 4 generations. Based on this information, choose the **correct** option showing the final number of meristematic cells and permanent cells formed from the zygote after last cycle of cell division.

	Number of permanent cells	Number of meristematic cells
(1)	16	16
(2)	8	16
(3)	72	39
(4)	32	8

(1) (1)
(2) (2)
(3) (3)
(4) (4)

123. Proximal end of the filament of stamen is attached to the

(1) Anther
(2) Placenta
(3) Connective
(4) Thalamus or petal

124. Match Column-I with Column-II and select the **correct** option.

	Column-I		Column-II
(I)	Henking	(a)	Coined the term linkage to describe physical association of genes on a chromosome
(II)	Alfred Sturtevant	(b)	Traced a specific nuclear structure all through spermatogenesis in a few insects.
(III)	T.H. Morgan	(c)	Rediscovered Mendel's results on the inheritance of characters.
(IV)	von Tschermak	(d)	Used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes

(1) (I)-(a), (II)-(d), (III)-(c), (IV)-(b)
(2) (I)-(b), (II)-(d), (III)-(a), (IV)-(c)
(3) (I)-(b), (II)-(a), (III)-(d), (IV)-(c)
(4) (I)-(c), (II)-(b), (III)-(a), (IV)-(d)

125. a accelerates abscission of leaves, flowers and fruits whereas b prevent this process.
Select the **correct** option to fill in the blanks a and b.

(1) **a** – Cytokinin, **b** – Auxin
(2) **a** – Gibberellin, **b** – Ethylene
(3) **a** – Abscisic acid, **b** – Ethylene
(4) **a** – Abscisic acid, **b** – Auxin

126. White kernel in coconut is

(1) Endosperm
(2) Haploid tissue
(3) Endocarp
(4) Product of zygote

127. Following are some events occurring in life cycle of more than 60% of angiosperms having true fruits.
(A) Formation of pollen tetrads via meiotic division.
(B) Reduction in water content and drying (10-15% moisture by mass) of fertilised ovule.
(C) Free nuclear division in triploid nuclei.
(D) Formation of an asymmetrical spindle within the product of microsporogenesis.
(E) Formation of male gametes via mitosis within the pollen tube.
(F) Formation of globular embryo.
Which of the following options shows the correct order of events that occur between pollen dispersal and seed dispersal?

(1) D → E → C → F → B
(2) E → B → C → F
(3) A → D → E → C → B
(4) E → C → F → B

CC-075
Vivek Kr. Singh
AaKash Medical/IIT-JEE/Foundations

128. A colourblind man marries a woman who is carrier of haemophilia. What is the probability of their offsprings having alleles for both colour blindness and haemophilia?

(1) 3/4

(2) 1/8

(3) 1/2

(4) 1/4

129. Multicarpellary apocarpous gynoecium is found in

(1) *Papaver*

(2) Chilli

(3) *Michelia*

(4) *Datura*

130. Read the following assertion and reason carefully.
Assertion (A): A bicelled pollen contains cells that appear dissimilar in size.
Reason (R) : Due to abundant food and large vacuole of generative cell the vegetative cell is pushed to the wall.
Choose the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true and (R) is false

(4) Both (A) and (R) are false

131. Read the following statement carefully.
The frequency of occurrence of colorblindness is ___A___ and ___B___ in males and females respectively.
Choose the **correct** option to fill in the blanks A and B respectively.

(1) 0.8% and 4%

(2) 8% and 0.4%

(3) 66.6% and 33.3%

(4) 0.75% and 0.25%

132. Mendel published his work on inheritance of characters in the year:

(1) 1822

(2) 1856

(3) 1863

(4) 1865

133. What percentage of progeny produced by self-pollination of a pea plant having genotype for intermediate size of starch grains would be hybrid for round seed shape?

(1) 50

(2) 25

(3) 75

(4) Zero

134. Select the statement that is **not** true about 2, 4-dichlorophenoxyacetic acid.

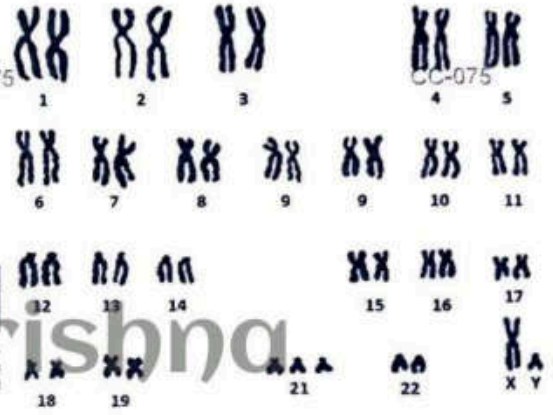
(1) It is a natural auxin produced in many plants

(2) It can induce the apical dominance

(3) It is widely used as herbicide

(4) It can be used extensively in agricultural and horticultural practices

135. Observe the following diagram.



Which of the following statements **correctly** depicts about the disorder developed due to the karyotype shown above?

(1) The underlying mechanism of the manifestation of this disorder is the effect of a gene on metabolic pathways which contribute towards different phenotypes.

(2) The individual affected with this disorder exhibit physical, psychomotor and retarded mental development.

(3) This disorder arises due to failure of cytokinesis after telophase stage of cell division.

(4) The individual developed from the zygote produced as a result of fertilisation of the egg (21 + X) by the sperm (22 + X).

CC-075 CC-075 CC-075 CC-075 Vivek Krishna Aakash Medical|IIT-JEE

ZOOLOGY

136. In an adult human, the enlarged end of penis that is covered by a loose fold of skin is homologous to tiny finger-like structure in female external genitalia which lies

(1) At the upper junction of two labia minora above the urethral opening

(2) At the upper junction of two labia majora below the urethral opening

(3) At the lower junction of labia minora below the vaginal opening

(4) At the lower junction of labia minora below the urethral opening but above the vaginal opening

137. In human females, the part of oviduct which has narrow lumen and joins the uterus, is lined by

(1) Ciliated cuboidal epithelium

(2) Brush bordered epithelium

(3) Ciliated columnar epithelium

(4) Simple squamous epithelium

138. An important characteristic that *Pleurobrachia* shares with *Pennatula* is the presence of

(1) Mesodermal layer between ectoderm and endoderm

(2) Property called metagenesis

(3) Skeleton composed of $CaCO_3$

(4) Both extracellular and intracellular digestion

139. Choose the option with correct set of animals that exhibit sexual dimorphism.

(1) *Pteropus*, *Taenia*

(2) *Pleurobrachia*, *Labeo*

(3) *Ancylostoma*, *Ornithorhynchus*

(4) *Hirudinaria*, *Anopheles*

140. In a non-pregnant human female, which of the following events is exclusively associated with the phase of ovarian cycle whose duration is fixed?

(1) Development of Graafian follicle

(2) Secretion of large amounts of progesterone

(3) Release of LH from the hypothalamus

(4) Release of estrogen from the developing ovarian follicles

141. Choose the character that is found in the members of the phylum that contains exclusively marine forms.

(1) Water canal system

(2) Jointed appendages

(3) Operculum

(4) Water vascular system

142. A proteinaceous hormone secreted from the female gonads in the latter phase of pregnancy

(1) Has similar mechanism of action as androgens

(2) Constricts the cervix and thickens the cervical mucus

(3) Aids in development of mammary glands

(4) Is named as relaxin

143. During extreme hot weather, adult frogs burrow themselves in mud and exhibit

(a) Buccopharyngeal respiration
(b) Cutaneous respiration
(c) High basal metabolic rate
(d) Locomotion for predation

Choose the **correct** option.

(1) (a), (b), (c) and (d)

(2) Only (a), (c) and (d)

(3) Only (b)

(4) Only (a) and (c)

144. In human females, movement of zygote/embryo through isthmus is facilitated by

(a) Coordinated beating of hair-like structures called cilia present on apical surface of specialised epithelial cells
(b) Contraction of uninucleated and branched muscles
(c) Peristalsis (contraction and relaxation of fusiform-shaped, unstriated muscle fibres)

Choose the **correct** option.

(1) (a) and (c)

(2) (a) and (b)

(3) Only (b)

(4) Only (a)

145. All of the following options are true w.r.t. *Rana tigrina*, **except**

(1) Presence of numerous finger-like folds in the inner wall of intestine

(2) Presence of paired tongue

(3) Muscular and myogenic heart

(4) Well organised organ of vision

CC-075 CC-075 CC-075 CC-075 CC-075 CC-075
Vivek Krishna
Aakash Medical | IIT-JEE | Foundations

146. In an experiment, a researcher labelled a female frog as sample 'P' and a male frog as sample 'Q'.
He observed notable differences between them, few of which are listed below.

	Sample P		Sample Q
(a)	Bidder's canal present	–	Bidder's canal absent
(b)	Gamete transport affected by cutting ureters	–	Gamete transport unaffected by cutting ureters
(c)	No functional connection of ovaries with the kidneys	–	Functional and structural connection of testes with kidneys
(d)	Presence of vasa efferentia	–	Absence of vasa efferentia

Choose the option representing only **incorrect** observations.

(1) (a), (b), (c) and (d)

(2) (a), (b) and (d)

(3) (b) and (c)

(4) (a), (b) and (c)

CC-075

147. Here given are the key classification events that signify the appearance of new traits that led to diversification of species and formation of major phyla that exist today. Arrange these classification events in evolutionary sequence from most recent to earliest.
a. Development of a cavity between the body wall and the gut wall lined by mesoderm
b. Type of symmetry where any plane passing through the central axis of the body divides the organism into identical halves
c. Multicellularity and arrangement of cells as loose aggregation.
d. A fluid-filled sac called amnion that encloses the embryo, cushioning it from physical stress.
e. Evolution of jaws
Choose the **correct** option.

(1) a → b → c → d → e

(2) d → e → b → a → c

(3) d → e → a → b → c

(4) c → b → a → e → d

148. Anterior portion of sperm head is covered by a cap-like structure that

(1) Possesses numerous mitochondria

(2) Is filled with enzymes that help in fertilisation

CC-075

(3) Contains diploid elongated nucleus

(4) Facilitates sperm motility due to the presence of flagella

149. Select the **incorrect** match.

(1)	*Spongilla*	–	Exhibits internal fertilization and indirect development
(2)	*Balanoglossus*	–	Exhibits external fertilisation and indirect development
(3)	*Pterophyllum*	–	Exhibits external fertilization and direct development
(4)	*Hippocampus*	–	Exhibits external fertilisation and indirect development

(1) (1)

(2) (2)

(3) (3)

(4) (4)

150. Choose the **incorrect** statement w.r.t. humans.

(1) Spermatogenesis and oogenesis require different optimum temperatures.

CC-075

(2) Testosterone causes a net increase in muscle protein but decreases production of red blood cell by reducing the stem cells proliferation

CC-075

(3) Spermatogonia present on the inside wall of seminiferous tubules undergo mitotic division

(4) The action of FSH on target cells is mediated through cyclic adenosine monophosphate after interaction with specific membrane bound receptors.

151. In *Rana tigrina*, all of the following are seen, **except**

(1) Paired olfactory lobes

(2) Paired diencephalon

(3) Paired optic lobes

(4) Paired fat bodies

152. How many of the following statements correctly distinguish between spermatogenesis and oogenesis in case of humans?
(i) Meiotic divisions required to produce each gamete
(ii) Total number of functional gametes produced after completion of meiotic division
(iii) Interruptions during meiotic divisions
(iv) Distribution of cytoplasm during cytokinesis
(v) Motility of functional gamete
Choose the **correct** option.

(1) Five

(2) Three

CC-075

(3) Four

CC-075

(4) Two

Vivek Kharat Aakash Medical|IIT-JEE|Foundations

153. Two types of circulatory system are depicted in figures A and B:

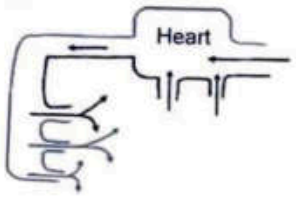


Fig. A

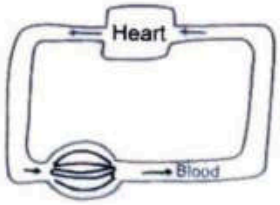


Fig. B

Match figures A and B with the resepective animals given below.

(i) *Scorpion*
(ii) *Pheretima*
(iii) *Pila*
(iv) *Salamandra*

Choose the correct option.

	Figure A	Figure B
(1)	(iii), (iv)	(i), (ii)
(2)	(ii), (iii)	(i), (iv)
(3)	(i), (iv)	(ii), (iii)
(4)	(i), (iii)	(ii), (iv)

CC-075

(1) (1)

(2) (2)

(3) (3)

(4) (4)

154. How many yellow coloured thin filamentous tubules that help in removal of excretory products from haemolymph, are present at the junction of foregut and midgut in cockroaches?

(1) 100-150

(2) 6-8

(3) Zero

(4) 12-13

155. Match the columns.

	Column I		Column II		Column III
(a)	Sea hare	(i)	Echinodermata	(1)	Calcareous endoskeleton
(b)	Sea lily	(ii)	Mollusca	(2)	File-like radula for rasping
(c)	Sea walnut	(iii)	Coelenterata	(3)	Cnidoblasts
(d)	Sea pen	(iv)	Ctenophora	(4)	External rows of ciliated comb plates for locomotion

CC-075 CC-075

Select the **correct** match.

(1) a-(i), (1) ; b-(ii), (2) ; c-(iii), (3) ; d-(iv), (4)

(2) a-(ii), (2) ; b-(iii), (1) ; c-(iv), (4) ; d-(i), (3)

(3) a-(ii), (2) ; b-(i), (1) ; c-(iv), (4) ; d-(iii), (3)

(4) a-(iii), (1) ; b-(ii), (4) ; c-(iv), (3) ; d-(ii), (2)

156. In a cockroach, how many walking legs and wings arise respectively from prothorax?

(1) 2 and 2

(2) 2 and 0

(3) 0 and 2

(4) 6 and 2

157. The cell in human female gonads which possesses only one sex chromosome and 22 autosomes is

(1) Second polar body

(2) Primary oocyte

(3) First polar body

(4) Granulosa cell

158. After implantation, finger-like projections appear on the trophoblast called _____ which are surrounded by the uterine tissue and maternal blood.
Fill in the blank by selecting the **correct** option.

(1) Clitoris

(2) Chorionic villi

(3) Yolk sac

(4) Umbilical cord

CC-075 CC-075

159. In female cockroach, anterior part of brood pouch contains opening for

(1) Gonopore, phallic gland, collaterial gland

(2) Gonopore, spermathecae, collaterial gland

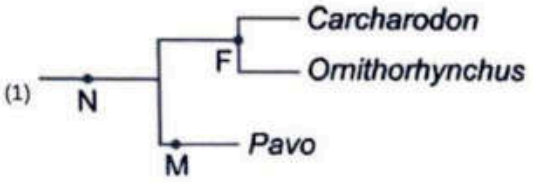
(3) Spermathecae, phallic gland, long tubules

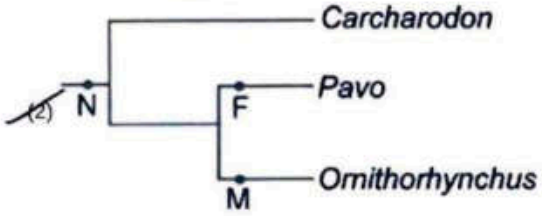
(4) Phallic gland, short tubules, long tubules

Vivek Kashyap
Aakash Medical|IIT-JEE|Foundations

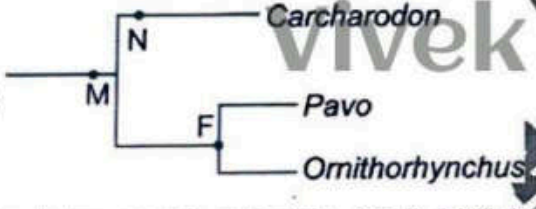
160. Consider three animals *i.e.*, *Carcharodon*, *Pavo*, *Ornithorhynchus* for the following body structures:
- Notochord (N)
- Feathers (F)
- Mammary glands (M)

Choose the cladogram that classifies the evolution of these animals most appropriately.

(1) 

(2) 

(3) 

(4) 

161. Given below are two statements, one is labelled as **Assertion(A)** and other is labelled as **Reason (R)**.
Assertion (A): In humans, despite of millions of sperms released per ejaculation, normally only one sperm can enter the secondary oocyte during fertilisation.
Reason (R) : After fertilisation, the acrosomal enzymes released from sperms completely degrade the zona pellucida which prevents the binding of additional sperm.
In the light of above statements, select the most appropriate answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are false

(3) (A) is true but (R) is false

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

162. Select the animal which is monoecious and exhibits external fertilisation.

(1) *Limulus*

(2) *Taenia*

(3) *Wuchereria*

(4) *Pheretima*

163. The yellowish fluid, secreted from mammary glands of mother during the initial days of lactation is very essential as it imparts immunity to the newborn. This fluid predominantly contains

(1) Antigens

(2) FSH

(3) Immunoglobin-A

(4) Luteinizing hormone

164. A researcher found 4 adult animals depicted as P, Q, R and S in the sea during an expedition. After careful study, he enlisted few features given below:-
- P is asymmetric, exhibits intracellular digestion, indirect development and can also reproduce by fragmentation.
- Q possesses feather-like gills but lacks calcareous shell and exhibits direct development.
- R is a gnathostome that possesses flat, diamond shaped body with enlarged pectoral fins that are fused with head and a sharp stinger in whip-like tail.
- S lacks cranium but possesses notochord towards dorsal side of body from head to tail.

Which among the following comparisons between P, Q, R and S is correct?

(1) R possesses gills with gill covers unlike S that lacks any specialised respiratory structure

(2) Q possesses blood capillaries unlike S in which blood exits from poorly developed blood vessels

(3) Both P and R exhibit internal fertilisation but P is hermaphrodite unlike R

(4) Both Q and S exhibit metameric segmentation but Q possesses gill slits unlike S

165. Development of heart is the first sign of growing human foetus. Heart originates from mesoderm which actually differentiates from

(1) Trophoblast

(2) Chorionic villi

(3) Inner cell mass

(4) Morula

CC-075 CC-075 CC-075 CC-075 CC-075
Vivek Kashyap Sir's Medical/IIT-JEE Coaching Classes

166. Observe the matches given below:

	Character	Frog	Human
a.	Cerebellum	Unpaired	Paired
b.	Renal portal system	Absent	Present
c.	Ventricle in heart	Paired	Paired
d.	Pons	Present	Present
e.	Hind limbs	Four digits	Five digits

Select the option representing **incorrect** matches.

(1) a, b, c and d only

(2) a, b, c, d and e

(3) a, b and d only

(4) a, c and d only

167. Which among the following has the same number of chromosomes in humans?

(1) Sertoli cells, secondary oocytes, granulosa cells

(2) Spermatogonium, zygote, primary oocyte

(3) Ova, spermatids, Leydig cells

(4) Leydig cells, secondary spermatocytes and spermatozoa

CC-075

168. All the organisms belonging to the class Amphibia

(1) Possess two forelimbs and two hindlimbs

(2) Possess body which is divisible only into head and trunk

(3) Possess mesodermal layer in their embryonic stage

(4) Are bisexual

169. Select the **odd** one w.r.t. function of estrogen in human females.

(1) Enlargement of external genitalia

(2) High pitch of voice

(3) Increased demineralisation of bones

(4) Marked proliferation of endometrium of uterus

170. All of the following are correct for *Ascaris*, *Aedes* and *Antedon*, **except**

(1) Exhibit similar level of body organisation

(2) Exhibit indirect development

(3) Possess complete digestive system

(4) Possess specialised excretory organs

171. Choose the homeothermic organism among the following which has the ability to fly as well as exhibits oviparity.

CC-075

(1) *Pteropus*

(2) *Pavo*

(3) *Culex*

(4) *Anopheles*

172. How many of the organisms listed below exhibit the type of symmetry which is seen in adult *Ophiura*?

(a) *Gorgonia*
(b) *Ichthyophis*
(c) *Pleurobrachia*
(d) *Aurelia*
(e) *Pterophyllum*

Select the **correct** option.

(1) Five

(2) Three

(3) Two

(4) Four

173. Choose the **correct** option w.r.t. *Aptenodytes*.

(1) Belongs to the division Tetrapoda

(2) Unable to maintain a constant body temperature

(3) Possesses flippers which are modified forelimbs

(4) Exhibits oviparity and external fertilisation

174. Read the following:

(a) Monoecious
(b) Internal fertilisation
(c) Lack of digestive system
(d) Presence of hooks and suckers

CC-075 CC-075

Above given features are true for

(1) *Fasciola*

(2) *Taenia*

(3) *Ancylostoma*

(4) *Nereis*

175. When frogs in grassland are transferred to dry soil, they change their colour from green to brownish-grey.

Select the **incorrect** option w.r.t. the above given statement.

(1) It occurs only during hibernation

(2) Adaptive colour change as a defense mechanism

(3) Involves dispersion and aggregation of pigments in cells present in the skin

(4) It is a physiological process vital for their survival

176. In human females, how many of the events mentioned in the box below occurs post fertilisation?

Syngamy, Parturition, Lactation, Formation of secondary oocyte, Gastrulation

Choose the **correct** option.

CC-075 CC-075

(1) Five

(2) Four

(3) Two

(4) Three

Vivek Aakash Medical/IIT-JEE/Foundations

177. Read the following statements w.r.t. humans.

Statement A: If the ovum is not fertilised, the corpus luteum in the ovary involutes and the level of ovarian hormones decreases.

Statement B: Estrogen in presence of progesterone has strong inhibitory effect on gonadotropins which are secreted from pars nervosa.

In the light of above statements, select the **correct** option.

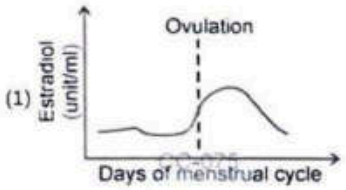
(1) Both statements A and B are correct

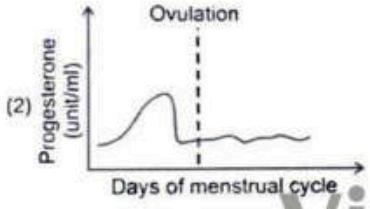
(2) Both statements A and B are incorrect

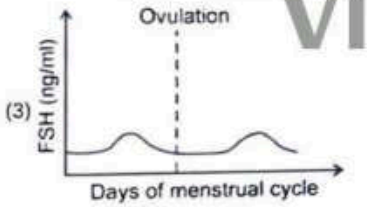
(3) Only statement A is correct

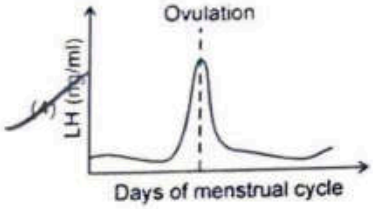
(4) Only statement B is correct

178. Which among the following graphs correctly represents the burst of secondary oocyte from mature follicle in human females?

(1)


(2)


(3)


(4)


CC-075

179. Among the below given options, select the organism that is exclusively marine and possesses true coelom.

(1) *Meandrina*

(2) *Bombyx*

(3) *Planaria*

(4) *Asterias*

180. Select the **incorrect** match w.r.t. gestation period and embryonic development in humans.

(1)	End of 2nd month of pregnancy	–	Development of limbs and digits
(2)	End of 12 weeks	–	First movement of foetus is observed
(3)	During the 5th month	–	Appearance of hair on the head
(4)	End of about 24th week	–	Eye-lids separate and eye lashes are formed

(1) (1)

(2) (2)

(3) (3)

(4) (4)

CC-075

CC-075

Vivek Krishna
Aakash Medical|IIT-JEE|Foundations